Chapter 1. Distributions and Inference for Categorical Data

Binomial and multinomial likelihood inference

Contents

1.1 Categorical response data **2**
- 1.1.1 Response–explanatory variable distinction . . . 2
- 1.1.2 Nominal–ordinal scale distinction . . . 2
- 1.1.3 Continuous–discrete variable distinction . . . 2
- 1.1.4 Quantitative–qualitative variable distinction . . . 2
- 1.1.5 Organization of this book . . . 2

1.2 Distributions for categorical data **2**
- 1.2.1 Binomial distribution . . . 3
- 1.2.2 Multinomial distribution . . . 3
- 1.2.3 Poisson distribution . . . 4
- 1.2.4 Overdispersion . . . 4
- 1.2.5 Connection between Poisson and multinomial distributions . . . 4

1.3 Statistical inference for categorical data **5**
- 1.3.1 Likelihood functions and maximum likelihood estimation . . . 5
- 1.3.2 Likelihood function and ML estimate for binomial parameter . . . 6
- 1.3.3 Wald–likelihood ratio–score test triad . . . 6
- 1.3.4 Constructing confidence intervals . . . 8

1.4 Statistical inference for binomial parameters **8**
- 1.4.1 Tests about a binomial parameter . . . 8
- 1.4.2 Confidence intervals for a binomial parameter . . . 8
- 1.4.3 Proportion of vegetarians example . . . 9

1.5 Statistical inference for multinomial parameters **9**
- 1.5.1 Estimation of multinomial parameters . . . 10
- 1.5.2 Pearson statistic for testing a specified multinomial . . . 10
- 1.5.3 Example: testing Mendel's theories . . . 10
- 1.5.4 Chi-squared theoretical justification . . . 10
- 1.5.5 Likelihood-ratio chi-squared . . . 11
- 1.5.6 Testing with estimated expected frequencies . . . 11

1.1 Categorical response data

Motivation. The measurement scale of a categorical response decides which models and summaries are meaningful.

A *categorical variable* has a measurement scale consisting of a set of categories (political philosophy: liberal / moderate / conservative; a mammogram diagnosis: normal, benign, suspicious, malignant).

1.1.1 Response–explanatory variable distinction

We distinguish a *response* variable from *explanatory* variables; these notes concern *categorical responses*, with explanatory variables of any type.

1.1.2 Nominal–ordinal scale distinction

Definition (Nominal, ordinal, interval)

(a) A categorical variable is *nominal* if its categories have no natural ordering (religious affiliation, mode of transport).
(b) It is *ordinal* if the categories are ordered but the distances between them are unknown (patient condition good/fair/serious/critical).
(c) An *interval* variable has numerical distances between any two values (blood pressure, income); it is a *ratio* variable if ratios are also meaningful.

The classification depends on how a variable is measured (education: public/private school, highest degree, years of schooling). In the hierarchy interval > ordinal > nominal, methods for a lower level can be used at a higher level (ignoring information) but not conversely.

1.1.3 Continuous–discrete variable distinction

All measurement is in practice discrete; the useful distinction is between variables taking many values and variables taking few values, which are the subject here.

1.1.4 Quantitative–qualitative variable distinction

Nominal variables are *qualitative*, interval variables *quantitative*. Ordinal variables sit in between, often with an underlying continuous variable; analysts exploit this by assigning *scores* or assuming a latent continuous distribution.

1.1.5 Organization of this book

The models replace normality by binomial, multinomial or Poisson distributions; the central families are *logistic regression* and *loglinear models*. Chapters 1–3 cover one- and two-way data; Chapters 4–9 generalized linear models; Chapters 10–13 clustered responses; Chapters 14–16 theory and history.

1.2 Distributions for categorical data

Motivation. Binomial, multinomial and Poisson play the role for counts that the normal plays for continuous responses.

1.2.1 Binomial distribution

Let $Y_1, \ldots, Y_n$ be independent Bernoulli trials with $P(Y_i = 1) = \pi$. Then $Y = \sum_i Y_i \sim \text{bin}(n, \pi)$:

$$p(y) = \binom{n}{y} \pi^y (1-\pi)^{n-y}, \qquad y = 0, 1, \ldots, n. \tag{1.1}$$

Proposition 1.1 (Binomial moments)

If $Y \sim \text{bin}(n, \pi)$ then $\mu = E(Y) = n\pi$, $\sigma^2 = \text{Var}(Y) = n\pi(1-\pi)$, and

$$\frac{E(Y-\mu)^3}{\sigma^3} = \frac{1-2\pi}{\sqrt{n\pi(1-\pi)}}.$$

In particular the distribution becomes symmetric (and, by the CLT, normal) as $n \to \infty$ for fixed π.

Proof.

Step 1. For one trial $E(Y_i) = E(Y_i^2) = \pi$, so $\text{Var}(Y_i) = \pi(1-\pi)$; by linearity and independence $E(Y) = n\pi$, $\text{Var}(Y) = n\pi(1-\pi)$.

Step 2. Third central moments of independent variables add (cumulants are additive), and $E(Y_i - \pi)^3 = (1-\pi)^3\pi - \pi^3(1-\pi) = \pi(1-\pi)(1-2\pi)$. Hence $E(Y-\mu)^3 = n\pi(1-\pi)(1-2\pi)$; dividing by $\sigma^3 = [n\pi(1-\pi)]^{3/2}$ gives the formula, which tends to 0. ■

Without independent, identical trials the binomial fails (e.g. sampling without replacement gives the hypergeometric, Section 3.5).

1.2.2 Multinomial distribution

Let each of n independent, identical trials fall in one of c categories, $y_{ij} = 1$ if trial i falls in category j, $n_j = \sum_i y_{ij}$ and $\pi_j = P(y_{ij} = 1)$. Then $(n_1, \ldots, n_c)$ is *multinomial*:

$$p(n_1, \ldots, n_{c-1}) = \frac{n!}{n_1!\, n_2! \cdots n_c!} \pi_1^{n_1} \pi_2^{n_2} \cdots \pi_c^{n_c}, \tag{1.2}$$

a $(c-1)$-dimensional distribution since $\sum_j n_j = n$; $c = 2$ gives the binomial.

Proposition 1.2 (Multinomial moments)

For the multinomial $(n, \boldsymbol{\pi})$ distribution,

$$E(n_j) = n\pi_j, \qquad \text{Var}(n_j) = n\pi_j(1-\pi_j), \qquad \text{Cov}(n_j, n_k) = -n\pi_j\pi_k \ (j \neq k),$$

and each n_j is marginally $\text{bin}(n, \pi_j)$.

Proof. [Supplement]

Step 1. n_j is a sum of n independent Bernoulli(π_j) indicators y_{ij}, so $n_j \sim \text{bin}(n, \pi_j)$ (Proposition 1.1).

Step 2. For $j \neq k$, $y_{ij}y_{ik} = 0$, so $\text{Cov}(y_{ij}, y_{ik}) = -\pi_j\pi_k$; summing over independent trials gives $\text{Cov}(n_j, n_k) = -n\pi_j\pi_k$. ■

In matrix form, $\text{Cov}(\mathbf{n}) = n[\text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}^\top]$.

1.2.3 Poisson distribution

Counts without a fixed upper limit are often modelled as *Poisson* with mean μ:

$$p(y) = \frac{e^{-\mu}\mu^y}{y!}, \qquad y = 0, 1, 2, \ldots \tag{1.3}$$

Its skewness is $1/\sqrt{\mu}$, so it approaches normality as $\mu \to \infty$.

Proposition 1.3 (Poisson moments and binomial limit)

(a) If $Y \sim \text{Poisson}(\mu)$ then $E(Y) = \text{Var}(Y) = \mu$.
(b) If $n \to \infty$ and $\pi \to 0$ with $n\pi \to \mu$, then $\text{bin}(n, \pi)$ probabilities converge to $\text{Poisson}(\mu)$ probabilities.

Proof. [Supplement]

Step (a). $E(Y) = \sum_{y\geq 1} e^{-\mu}\mu^y/(y-1)! = \mu$ and $E[Y(Y-1)] = \sum_{y\geq 2} e^{-\mu}\mu^y/(y-2)! = \mu^2$, so $\text{Var}(Y) = \mu^2 + \mu - \mu^2 = \mu$.

Step (b). $\binom{n}{y}\pi^y(1-\pi)^{n-y} = \frac{(n\pi)^y}{y!} \cdot \frac{n(n-1)\cdots(n-y+1)}{n^y} \cdot (1-\pi)^{n-y}$. For fixed y the middle factor $\to 1$, and $(1-\pi)^{n-y} = \exp\{-n\pi + O(n\pi^2)\}(1-\pi)^{-y} \to e^{-\mu}$. ■

1.2.4 Overdispersion

Real counts often vary more than the binomial or Poisson allows (*overdispersion*), because the success probability or rate itself varies with unmeasured factors.

Proposition 1.4 (Mixing creates overdispersion)

Let Y given λ have mean $E(Y \mid \lambda)$ and variance $\text{Var}(Y \mid \lambda)$, where λ is random. Then

$$E(Y) = E[E(Y \mid \lambda)], \qquad \text{Var}(Y) = E[\text{Var}(Y \mid \lambda)] + \text{Var}[E(Y \mid \lambda)].$$

If $Y \mid \lambda \sim \text{Poisson}(\lambda)$ with $E(\lambda) = \mu$, then $E(Y) = \mu$ and $\text{Var}(Y) = \mu + \text{Var}(\lambda) > \mu$ whenever λ is nondegenerate.

Proof. [Supplement]

Step 1. The first identity is the tower property. For the second, $\text{Var}(Y) = E[\text{Var}(Y \mid \lambda) + E(Y \mid \lambda)^2] - (E[E(Y \mid \lambda)])^2 = E[\text{Var}(Y \mid \lambda)] + \text{Var}[E(Y \mid \lambda)]$.

Step 2. In the Poisson case $E(Y \mid \lambda) = \text{Var}(Y \mid \lambda) = \lambda$. ■

Mixtures of binomials likewise give extra-binomial variation (negative binomial, Section 4.3.4; random effects, Chapters 12–13).

1.2.5 Connection between Poisson and multinomial distributions

Theorem 1.5 (Poisson conditioned on the total is multinomial)

Let $Y_1, \ldots, Y_c$ be independent Poisson with means $\mu_1, \ldots, \mu_c$. Then $n = \sum_i Y_i \sim \text{Poisson}(\sum_i \mu_i)$, and conditional on n,

$$(Y_1, \ldots, Y_c) \mid \textstyle\sum_i Y_i = n \;\sim\; \text{multinomial}(n, \{\pi_i\}), \qquad \pi_i = \mu_i / \textstyle\sum_j \mu_j.$$

Proof.

Step 1. The product of the pgfs $e^{\mu_i(s-1)}$ is $e^{(\sum \mu_i)(s-1)}$, the pgf of Poisson$(\sum \mu_i)$.

Step 2. For $\sum n_i = n$, the conditional probability is

$$\frac{\prod_i e^{-\mu_i}\mu_i^{n_i}/n_i!}{e^{-\sum_j \mu_j}(\sum_j \mu_j)^n/n!} = \frac{n!}{\prod_i n_i!}\prod_i\Big(\frac{\mu_i}{\sum_j \mu_j}\Big)^{n_i},$$

using $\sum_i n_i = n$ to distribute $(\sum_j \mu_j)^n$; this is (1.2). ∎

Since the Poisson and multinomial likelihoods share the kernel in $\{\pi_i\}$, the two models usually give the same estimates.

1.3 Statistical inference for categorical data

Motivation. Maximum likelihood and the Wald, likelihood-ratio and score tests, with the intervals obtained by inverting them, form the toolkit for the whole course.

1.3.1 Likelihood functions and maximum likelihood estimation

Definition (Likelihood, ML estimate, information)

The *likelihood* $\ell(\beta)$ is the probability of the observed data as a function of β, and $L(\beta) = \log \ell(\beta)$. The *ML estimate* $\hat\beta$ maximizes ℓ, typically solving $\partial L/\partial\beta = 0$. The *information matrix* has (j,k) element

$$-E\Big(\frac{\partial^2 L(\boldsymbol\beta)}{\partial\beta_j\,\partial\beta_k}\Big). \tag{1.4}$$

Under regularity conditions $\mathrm{Cov}(\hat{\boldsymbol\beta})$ is asymptotically the inverse information; greater curvature of L means smaller SEs.

Lemma 1.6 (Score identities)

Let $f(y;\beta)$ be a density or pmf such that differentiation under $\int dy$ (or $\sum_y$) is allowed twice. For one observation let $u(\beta) = \partial \log f(Y;\beta)/\partial\beta$. Then

$$E_\beta[u(\beta)] = 0, \qquad \mathrm{Var}_\beta[u(\beta)] = -E_\beta\Big[\frac{\partial^2 \log f(Y;\beta)}{\partial\beta^2}\Big] =: i(\beta).$$

Proof. [Supplement]

Step 1. Differentiate $\int f\,dy = 1$: $0 = \int (\partial_\beta \log f)\, f\, dy = E[u]$.

Step 2. Differentiate again: $0 = \int [(\partial_\beta^2 \log f) f + (\partial_\beta \log f)^2 f]\,dy$, so $\mathrm{Var}(u) = E[u^2] = -E[\partial_\beta^2 \log f]$. ∎

Theorem 1.7 (Asymptotic normality of the ML estimator)

Let $Y_1,\dots,Y_n$ be i.i.d. $f(y;\beta_0)$ with β_0 interior to the parameter space, satisfying Lemma 1.6, $0 < i(\beta_0) < \infty$, and $|\partial_\beta^3 \log f(y;\beta)| \le M(y)$ near β_0 with $E\,M(Y) < \infty$. If $\hat\beta$ is a consistent root of the likelihood equation, then

$$\sqrt{n}(\hat\beta - \beta_0) \xrightarrow{d} N\big(0,\, 1/i(\beta_0)\big).$$

For a vector $\boldsymbol{\beta}$ the same holds with $i(\boldsymbol{\beta}_0)$ the per-observation information matrix: $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, i(\boldsymbol{\beta}_0)^{-1})$, and in fact $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = i(\boldsymbol{\beta}_0)^{-1}\, n^{-1/2} \nabla L(\boldsymbol{\beta}_0) + o_p(1)$.

Proof. [Supplement]

Step 1. Expanding $0 = L'(\hat{\beta}) = L'(\beta_0) + (\hat{\beta} - \beta_0)L''(\beta_0) + \frac{1}{2}(\hat{\beta} - \beta_0)^2 L'''(\beta^*)$, with β^* between $\hat{\beta}$ and β_0,

$$\sqrt{n}(\hat{\beta} - \beta_0) = \frac{n^{-1/2} L'(\beta_0)}{-n^{-1} L''(\beta_0) - \frac{1}{2}(\hat{\beta} - \beta_0)\, n^{-1} L'''(\beta^*)}.$$

Step 2. $L'(\beta_0)$ is a sum of i.i.d. terms with mean 0 and variance $i(\beta_0)$ (Lemma 1.6), so $n^{-1/2} L'(\beta_0) \xrightarrow{d} N(0, i(\beta_0))$.

Step 3. $-n^{-1} L''(\beta_0) \xrightarrow{p} i(\beta_0)$ (WLLN), and with probability $\to 1$, $|n^{-1} L'''(\beta^*)| \le n^{-1} \sum_i M(Y_i) = O_p(1)$, which times $\hat{\beta} - \beta_0 = o_p(1)$ vanishes. So the denominator $\xrightarrow{p} i(\beta_0)$.

Step 4. Slutsky gives $N(0, i(\beta_0))/i(\beta_0) = N(0, 1/i(\beta_0))$ and the stated representation. The vector case replaces L'' by the Hessian and division by matrix inversion. ■

1.3.2 Likelihood function and ML estimate for binomial parameter

Constants such as $\binom{n}{y}$ do not affect the maximizer; only the *kernel* matters.

Proposition 1.8 (Binomial ML estimate and information)

For y successes in n trials, the log likelihood, score and information are

$$L(\pi) = y \log \pi + (n - y) \log(1 - \pi), \tag{1.5}$$

$$\frac{\partial L}{\partial \pi} = \frac{y}{\pi} - \frac{n - y}{1 - \pi} = \frac{y - n\pi}{\pi(1 - \pi)}, \tag{1.6}$$

$$-E\Big(\frac{\partial^2 L}{\partial \pi^2}\Big) = \frac{n}{\pi(1 - \pi)}. \tag{1.7}$$

The ML estimate is $\hat{\pi} = y/n$, with $E(\hat{\pi}) = \pi$ and $\sigma(\hat{\pi}) = \sqrt{\pi(1 - \pi)/n}$, the inverse square root of the information.

Proof.

Step 1. (1.6) vanishes iff $\pi = y/n$, and $\partial^2 L/\partial \pi^2 = -y/\pi^2 - (n - y)/(1 - \pi)^2 < 0$, so y/n is the global maximizer (for $y = 0$ or n, L is monotone with maximum at the boundary, again y/n).

Step 2. With $E(Y) = n\pi$: $E[Y/\pi^2 + (n - Y)/(1 - \pi)^2] = n/[\pi(1 - \pi)]$; and $E(\hat{\pi}) = \pi$, $\mathrm{Var}(\hat{\pi}) = \pi(1 - \pi)/n$. ■

1.3.3 Wald–likelihood ratio–score test triad

For $H_0 \colon \beta = \beta_0$ write $L_0 = L(\beta_0)$ and $L_1 = L(\hat{\beta})$.

Definition (The three likelihood-based statistics)

(a) *Wald*: $z_W = (\hat{\beta} - \beta_0)/\mathrm{SE}$ with SE evaluated at $\hat{\beta}$; for a vector, $W = (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)^\top [\widehat{\mathrm{Cov}}(\hat{\boldsymbol{\beta}})]^{-1} (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)$.
(b) *Likelihood ratio (LR)*: $\Lambda = \ell_0/\ell_1 \le 1$ and $-2 \log \Lambda = -2(L_0 - L_1)$.

(c) *Score*: with $u(\beta_0) = \partial L/\partial\beta|_{\beta_0}$ and $\iota(\beta_0) = -E[\partial^2 L/\partial\beta^2]|_{\beta_0}$, $z_S = u(\beta_0)/\iota(\beta_0)^{1/2}$; in the vector case $\mathbf{u}^\top \iota^{-1}\mathbf{u}$ at $\boldsymbol{\beta}_0$.

Wald uses L at $\hat\beta$, score its slope and curvature at β_0, and LR both values (Figure 1).

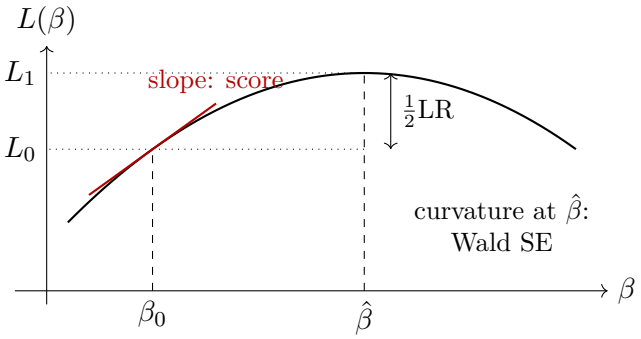


Figure 1: Information used by the three tests of $H_0\colon \beta = \beta_0$.

Lemma 1.9 (Normal quadratic forms)

(a) If $\mathbf{z} \sim N_q(\mathbf{0}, \Sigma)$ with Σ nonsingular, then $\mathbf{z}^\top\Sigma^{-1}\mathbf{z} \sim \chi^2_q$.
(b) If $\mathbf{z} \sim N_c(\mathbf{0}, P)$ with P symmetric idempotent of rank r, then $\mathbf{z}^\top\mathbf{z} \sim \chi^2_r$.

Proof. [Supplement]

Step (a). $\mathbf{w} = \Sigma^{-1/2}\mathbf{z} \sim N_q(\mathbf{0}, I)$ and $\mathbf{z}^\top\Sigma^{-1}\mathbf{z} = \sum_{k=1}^q w_k^2$.

Step (b). $P = Q\Delta Q^\top$ with Q orthogonal and Δ diagonal with r ones and zeros elsewhere (eigenvalues of an idempotent are 0 or 1). Then $\mathbf{w} = Q^\top\mathbf{z} \sim N(\mathbf{0}, \Delta)$, so $\mathbf{z}^\top\mathbf{z} = \mathbf{w}^\top\mathbf{w} = \sum_{k\le r} w_k^2 \sim \chi^2_r$. ■

Theorem 1.10 (Asymptotic null distributions of the triad)

Under the conditions of Theorem 1.7 and $H_0\colon \boldsymbol{\beta} = \boldsymbol{\beta}_0$ ($\boldsymbol{\beta} \in \mathbb{R}^q$), the Wald, LR and score statistics each converge in distribution to χ^2_q, and the difference between any two of them converges in probability to 0. For $q = 1$, z_W and z_S are asymptotically $N(0,1)$.

Proof. [Supplement] Write $\iota_1 = i(\boldsymbol{\beta}_0)$ and $\mathbf{s}_n = n^{-1/2}\nabla L(\boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \iota_1)$ (CLT, Lemma 1.6).

Step 1 (score). The score statistic is $\nabla L^\top (n\iota_1)^{-1}\nabla L = \mathbf{s}_n^\top \iota_1^{-1}\mathbf{s}_n \xrightarrow{d} \chi^2_q$ by Lemma 1.9(a).

Step 2 (Wald). By Theorem 1.7, $\mathbf{z}_n = \sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = \iota_1^{-1}\mathbf{s}_n + o_p(1)$, and the Wald statistic is $\mathbf{z}_n^\top \hat\iota_1 \mathbf{z}_n$ with $\hat\iota_1 \xrightarrow{p} \iota_1$; so it equals $\mathbf{s}_n^\top\iota_1^{-1}\mathbf{s}_n + o_p(1)$.

Step 3 (LR). Expanding about $\hat{\boldsymbol{\beta}}$, where $\nabla L = \mathbf{0}$, $L(\boldsymbol{\beta}_0) = L(\hat{\boldsymbol{\beta}}) + \frac{1}{2}(\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}})^\top \nabla^2 L(\tilde{\boldsymbol{\beta}})(\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}})$ with $\tilde{\boldsymbol{\beta}}$ on the segment. Hence $-2(L_0 - L_1) = \mathbf{z}_n^\top[-n^{-1}\nabla^2 L(\tilde{\boldsymbol{\beta}})]\mathbf{z}_n = \mathbf{z}_n^\top \iota_1 \mathbf{z}_n + o_p(1)$ (WLLN and the third-derivative bound), the Wald statistic plus $o_p(1)$.

Step 4. All three are $\mathbf{s}_n^\top\iota_1^{-1}\mathbf{s}_n + o_p(1) \xrightarrow{d} \chi^2_q$. For $q = 1$, $z_S = \mathbf{s}_n/\sqrt{\iota_1} \xrightarrow{d} N(0,1)$ and $z_W = z_S + o_p(1)$. ■

For composite H_0, Wilks' theorem gives $-2\log\Lambda \xrightarrow{d} \chi^2_{\mathrm{df}}$, df being the difference in parameter-space dimensions (Theorem 1.18 proves this for the multinomial). For small to moderate n the LR test is usually more reliable than Wald.

1.3.4 Constructing confidence intervals

Inverting a test gives an interval: the 95% interval is the set of β_0 whose test has $P > 0.05$.

(a) *Wald interval*: $\hat{\beta} \pm z_{\alpha/2}\mathrm{SE}$.

(b) *LR interval*: $\{\beta_0 : -2[L(\beta_0) - L(\hat{\beta})] < \chi^2_1(\alpha)\}$, where $\chi^2_1(\alpha) = z^2_{\alpha/2}$.

(c) *Score interval*: $\{\beta_0 : |z_S(\beta_0)| < z_{\alpha/2}\}$.

If L is exactly parabolic (normal $\hat{\beta}$) the three coincide; with small categorical samples they can disagree markedly. The LR interval is preferable for small to moderate n; Wald is the most common.

1.4 Statistical inference for binomial parameters

Motivation. Near the boundary the three large-sample methods already behave very differently for the binomial.

1.4.1 Tests about a binomial parameter

Proposition 1.11 (Binomial test statistics)

For $H_0\colon \pi = \pi_0$ with $\hat{\pi} = y/n$:

$$z_W = \frac{\hat{\pi} - \pi_0}{\sqrt{\hat{\pi}(1-\hat{\pi})/n}}, \tag{1.8}$$

$$z_S = \frac{\hat{\pi} - \pi_0}{\sqrt{\pi_0(1-\pi_0)/n}}, \tag{1.9}$$

$$-2(L_0 - L_1) = 2\sum \text{observed} \log\frac{\text{observed}}{\text{fitted}} = 2\Big[y\log\frac{y}{n\pi_0} + (n-y)\log\frac{n-y}{n-n\pi_0}\Big]. \tag{1.10}$$

Each is asymptotically $N(0,1)$ or χ^2_1 under H_0.

Proof.

Step 1. Wald uses $\mathrm{SE} = \sqrt{\hat{\pi}(1-\hat{\pi})/n}$ (Proposition 1.8 at $\pi = \hat{\pi}$).

Step 2. Score: by (1.6)–(1.7), $u(\pi_0)/\iota(\pi_0)^{1/2} = \frac{y - n\pi_0}{\pi_0(1-\pi_0)}\sqrt{\frac{\pi_0(1-\pi_0)}{n}} = \frac{y-n\pi_0}{\sqrt{n\pi_0(1-\pi_0)}}$, which is (1.9).

Step 3. LR: $-2(L_0 - L_1) = 2[y\log(\hat{\pi}/\pi_0) + (n-y)\log((1-\hat{\pi})/(1-\pi_0))]$, which is (1.10). The asymptotics are Theorem 1.10 with $q = 1$. ∎

The score statistic uses the null SE, and its null distribution is closer to $N(0,1)$.

1.4.2 Confidence intervals for a binomial parameter

Inverting the Wald test gives

$$\hat{\pi} \pm z_{\alpha/2}\sqrt{\hat{\pi}(1-\hat{\pi})/n}, \tag{1.11}$$

whose coverage is usually below nominal, much below for π near 0 or 1 (adding about 2 successes and 2 failures helps greatly).

Proposition 1.12 (Wilson score interval)

The score interval $\{\pi_0 : |\hat\pi - \pi_0| < z\sqrt{\pi_0(1-\pi_0)/n}\}$, $z = z_{\alpha/2}$, has endpoints

$$\hat\pi\Big(\frac{n}{n+z^2}\Big) + \frac{1}{2}\Big(\frac{z^2}{n+z^2}\Big) \;\pm\; z\sqrt{\frac{1}{n+z^2}\Big[\hat\pi(1-\hat\pi)\Big(\frac{n}{n+z^2}\Big) + \frac{1}{2}\cdot\frac{1}{2}\Big(\frac{z^2}{n+z^2}\Big)\Big]}.$$

Its midpoint is $(y + z^2/2)/(n + z^2)$, the sample proportion after adding $z^2/2$ successes and $z^2/2$ failures.

Proof. [Supplement]

Step 1. The set $|z_S| < z$ is where $(\hat\pi - \pi_0)^2 - z^2\pi_0(1-\pi_0)/n < 0$, i.e. between the roots of $(1 + z^2/n)\pi_0^2 - (2\hat\pi + z^2/n)\pi_0 + \hat\pi^2 = 0$.

Step 2. Multiplying by n, the quadratic formula gives $\pi_0 = \big[n\hat\pi + z^2/2 \pm z\sqrt{n\hat\pi(1-\hat\pi) + z^2/4}\big]/(n + z^2)$; dividing numerator terms by $n + z^2$ gives the stated centre and half-width. ■

The squared coefficient of z averages the variance of a proportion at $\hat\pi$ and at $1/2$, with sample size $n + z^2$. Wilson performs much better than Wald.

1.4.3 Proportion of vegetarians example

Example 1.13 (Proportion of vegetarians)

Of $n = 25$ students, $y = 0$ were vegetarian, so $\hat\pi = 0$.
(a) Wald: $(0, 0)$, useless at the boundary.
(b) Score: $(0.0,\ 0.133)$; e.g. $\pi_0 = 0.10$ gives $z_S = -1.67$ (inside).
(c) LR: $L(\pi) = 25\log(1-\pi)$, $L(\hat\pi) = 0$; the upper bound solves $-50\log(1-\pi_0) = 3.84$, giving $(0.0,\ 0.074)$.
The methods differ greatly because $\hat\pi$ is highly skewed near 0.

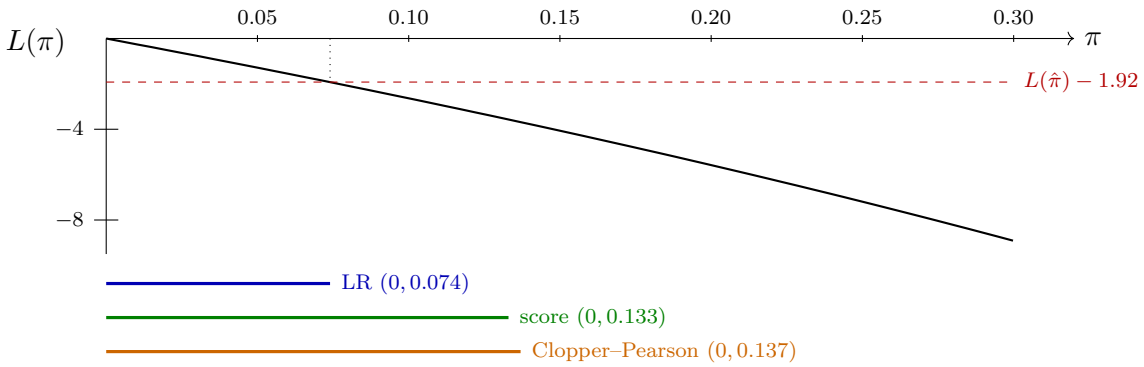


Figure 2: Binomial log likelihood for $y = 0$, $n = 25$; the LR interval is where L is within $3.84/2$ of its maximum.

1.5 Statistical inference for multinomial parameters

Motivation. Goodness of fit of a multinomial leads to Pearson's X^2 and the likelihood-ratio G^2, the workhorse statistics of the subject.

1.5.1 Estimation of multinomial parameters

Proposition 1.14 (Multinomial ML estimates)

The multinomial kernel

$$\prod_j \pi_j^{n_j}, \qquad \pi_j \geq 0, \quad \sum_j \pi_j = 1, \tag{1.12}$$

is maximized uniquely by the sample proportions $\hat{\pi}_j = p_j = n_j/n$.

Proof.

Step 1 (likelihood equations). With $\pi_c = 1 - \sum_{j<c} \pi_j$, $L = \sum_j n_j \log \pi_j$ has $\partial L/\partial \pi_j = n_j/\pi_j - n_c/\pi_c = 0$, $j < c$. So $\hat{\pi}_j/\hat{\pi}_c = n_j/n_c$; summing, $\hat{\pi}_c = n_c/n$ and $\hat{\pi}_j = n_j/n$.

Step 2 (global maximum). [Supplement] For any admissible $\boldsymbol{\pi}$ (sum over j with $n_j > 0$), Jensen's inequality gives

$$L(\boldsymbol{\pi}) - L(\mathbf{p}) = n \sum_j p_j \log \frac{\pi_j}{p_j} \leq n \log \sum_j p_j \frac{\pi_j}{p_j} \leq n \log 1 = 0,$$

with equality iff $\pi_j = p_j$ for all j. ■

1.5.2 Pearson statistic for testing a specified multinomial

For $H_0 \colon \pi_j = \pi_{j0}$ (Pearson, 1900), with *expected frequencies* $\mu_j = n\pi_{j0}$,

$$X^2 = \sum_j \frac{(n_j - \mu_j)^2}{\mu_j}, \tag{1.13}$$

and for large n, $X^2 \approx \chi^2_{c-1}$ (Theorem 1.16).

1.5.3 Example: testing Mendel's theories

Example 1.15 (Mendel's peas)

Mendel predicted 75% yellow seeds; of $n = 8023$, 6022 were yellow. Then $\mu_1 = 6017.25$, $X^2 = 0.015$ (df $= 1$), $P = 0.90$. Fisher (1936), summing independent statistics over Mendel's experiments, got $X^2 = 42$ with df $= 84$ (mean 84, SD 13): left-tail P is tiny, so the fit was *too good*, suggesting adjusted data.

1.5.4 Chi-squared theoretical justification

Theorem 1.16 (Asymptotic null distribution of Pearson's X^2)

Under $H_0 \colon \boldsymbol{\pi} = \boldsymbol{\pi}_0$ with all $\pi_{j0} > 0$, $X^2 \xrightarrow{d} \chi^2_{c-1}$ as $n \to \infty$. Moreover X^2 is the score statistic for H_0.

Proof. [Supplement] Let $\hat{\boldsymbol{\pi}} = (p_1, \ldots, p_{c-1})^\top$, $\boldsymbol{\pi}_0 = (\pi_{10}, \ldots, \pi_{c-1,0})^\top$ and $\Sigma_0 = \operatorname{diag}(\boldsymbol{\pi}_0) - \boldsymbol{\pi}_0 \boldsymbol{\pi}_0^\top$.

Step 1 (CLT). $\hat{\boldsymbol{\pi}}$ is a mean of i.i.d. indicator vectors with covariance Σ_0 (Proposition 1.2), so $\sqrt{n}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}_0) \xrightarrow{d} N_{c-1}(\mathbf{0}, \Sigma_0)$.

Step 2 (inverse). $\Sigma_0^{-1} = \operatorname{diag}(1/\pi_{j0}) + \mathbf{1}\mathbf{1}^\top/\pi_{c0}$: using $\boldsymbol{\pi}_0^\top \mathbf{1} = 1 - \pi_{c0}$, the product is $I + \boldsymbol{\pi}_0 \mathbf{1}^\top [1 - \pi_{c0} - (1 - \pi_{c0})]/\pi_{c0} = I$.

Step 3 (algebra). With $d_j = p_j - \pi_{j0}$, $j = 1, \dots, c$, and $d_c = -\sum_{j<c} d_j$,

$$n\,\mathbf{d}^\top \Sigma_0^{-1}\mathbf{d} = n\Big[\sum_{j<c}\frac{d_j^2}{\pi_{j0}} + \frac{(\sum_{j<c} d_j)^2}{\pi_{c0}}\Big] = n\sum_{j=1}^{c}\frac{d_j^2}{\pi_{j0}} = X^2. \tag{1.14}$$

Step 4 (limit). By continuous mapping and Lemma 1.9(a), $X^2 \xrightarrow{d} \chi^2_{c-1}$.

Step 5 (score test). By Proposition 1.14, Step 1, the score at $\boldsymbol{\pi}_0$ is $u_j = n(d_j/\pi_{j0} - d_c/\pi_{c0}) = n(\Sigma_0^{-1}\mathbf{d})_j$, and the information is $n\Sigma_0^{-1}$. So the score statistic is $\mathbf{u}^\top(n\Sigma_0^{-1})^{-1}\mathbf{u} = n\mathbf{d}^\top\Sigma_0^{-1}\mathbf{d} = X^2$. ∎

Heuristically (Fisher), independent Poisson n_j give $\sum_j (n_j - \mu_j)^2/\mu_j \approx \chi^2_c$, and conditioning on $\sum n_j = n$ (Theorem 1.5) loses one df. For $c = 2$, $X^2 = z_S^2$ with z_S from (1.9).

1.5.5 Likelihood-ratio chi-squared

The kernel (1.12) is maximized at $\boldsymbol{\pi}_0$ under H_0 and at p_j in general, so

$$G^2 = -2\log\Lambda = 2\sum_j n_j \log\frac{n_j}{n\pi_{j0}}, \tag{1.15}$$

of the form (1.10), with $G^2 \xrightarrow{d} \chi^2_{c-1}$ under H_0 (Theorem 1.10, $q = c - 1$).

Theorem 1.17 (Asymptotic equivalence of X^2 and G^2)

Let μ_j be (possibly estimated) expected frequencies with $\sum_j \mu_j = n$ and $(n_j - \mu_j)/\mu_j = O_p(n^{-1/2})$ for each j, as holds under H_0. Then $G^2 - X^2 \xrightarrow{p} 0$; in particular $G^2 \xrightarrow{d} \chi^2_{c-1}$ under $H_0\colon \boldsymbol{\pi} = \boldsymbol{\pi}_0$.

Proof. [Supplement]

Step 1. Write $n_j = \mu_j(1 + x_j)$, $x_j = O_p(n^{-1/2})$. Since $(1+x)\log(1+x) = x + x^2/2 + R(x)$ with $|R(x)| \le C|x|^3$ for $|x| \le 1/2$,

$$G^2 = 2\sum_j \mu_j x_j + \sum_j \mu_j x_j^2 + 2\sum_j \mu_j R(x_j).$$

Step 2. $\sum_j \mu_j x_j = \sum_j (n_j - \mu_j) = 0$, $\sum_j \mu_j x_j^2 = X^2$, and $\mu_j R(x_j) = O(n)O_p(n^{-3/2}) \xrightarrow{p} 0$. Slutsky with Theorem 1.16 gives the limit. ∎

Under H_a both grow like n but need not be close. X^2 usually approaches chi-squared faster; G^2 is often poor when $n/c < 5$.

1.5.6 Testing with estimated expected frequencies

Often $\pi_{j0} = \pi_j(\boldsymbol{\theta})$ with $\dim\boldsymbol{\theta} = p$ unknown; then $\hat\mu_j = n\pi_j(\hat{\boldsymbol{\theta}})$, and estimation reduces the df.

Theorem 1.18 (Degrees of freedom with estimated parameters)

Assume $\boldsymbol{\pi}(\boldsymbol{\theta})$ has continuous second derivatives, $\pi_j(\boldsymbol{\theta}_0) > 0$, the $c \times p$ Jacobian $\partial\boldsymbol{\pi}/\partial\boldsymbol{\theta}^\top$ has rank p at $\boldsymbol{\theta}_0$, and $\hat{\boldsymbol{\theta}}$ is a consistent ML estimate. If $H_0\colon \boldsymbol{\pi} = \boldsymbol{\pi}(\boldsymbol{\theta})$ holds, then

$$X^2 = \sum_j \frac{(n_j - \hat{\mu}_j)^2}{\hat{\mu}_j} \xrightarrow{d} \chi^2_{c-1-p}, \qquad \text{and } G^2 - X^2 \xrightarrow{p} 0.$$

Proof. [Supplement] Let $\boldsymbol{\pi}_0 = \boldsymbol{\pi}(\boldsymbol{\theta}_0)$, $D = \operatorname{diag}(\boldsymbol{\pi}_0)$, $\mathbf{s} = (\sqrt{\pi_{10}}, \ldots, \sqrt{\pi_{c0}})^\top$ (a unit vector), and $A = D^{-1/2}\,\partial\boldsymbol{\pi}/\partial\boldsymbol{\theta}^\top$ at $\boldsymbol{\theta}_0$ ($c \times p$, rank p).

Step 1. Since $\sum_j \pi_j(\boldsymbol{\theta}) = 1$, $\mathbf{s}^\top A = \sum_j \partial\pi_j/\partial\boldsymbol{\theta}^\top = \mathbf{0}$.

Step 2. With $\mathbf{p} = (p_1, \ldots, p_c)^\top$ and $\operatorname{Cov}(\sqrt{n}\,\mathbf{p}) = D - \boldsymbol{\pi}_0\boldsymbol{\pi}_0^\top$, the CLT gives $\mathbf{z}_n := \sqrt{n}\,D^{-1/2}(\mathbf{p} - \boldsymbol{\pi}_0) \xrightarrow{d} \mathbf{z} \sim N_c(\mathbf{0},\, I - \mathbf{s}\mathbf{s}^\top)$.

Step 3. $L(\boldsymbol{\theta}) = n\sum_j p_j \log \pi_j(\boldsymbol{\theta})$ has score at $\boldsymbol{\theta}_0$

$$\nabla L = n\,A^\top D^{-1/2}\mathbf{p} = n\,A^\top D^{-1/2}(\mathbf{p} - \boldsymbol{\pi}_0) = \sqrt{n}\,A^\top \mathbf{z}_n,$$

using $A^\top D^{-1/2}\boldsymbol{\pi}_0 = A^\top \mathbf{s} = \mathbf{0}$. The per-observation information is $\sum_j (\partial\pi_j/\partial\boldsymbol{\theta})(\partial\pi_j/\partial\boldsymbol{\theta})^\top/\pi_j = A^\top A$, so by Theorem 1.7 $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = (A^\top A)^{-1}A^\top \mathbf{z}_n + o_p(1)$.

Step 4. By the delta method, $\sqrt{n}\,D^{-1/2}(\boldsymbol{\pi}(\hat{\boldsymbol{\theta}}) - \boldsymbol{\pi}_0) = A\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + o_p(1)$. So with $H = A(A^\top A)^{-1}A^\top$,

$$\mathbf{e}_n := \sqrt{n}\,D^{-1/2}(\mathbf{p} - \boldsymbol{\pi}(\hat{\boldsymbol{\theta}})) = (I - H)\mathbf{z}_n + o_p(1) \xrightarrow{d} (I - H)\mathbf{z}.$$

Step 5. X^2 differs from $\mathbf{e}_n^\top \mathbf{e}_n$ only by $\pi_j(\hat{\boldsymbol{\theta}})$ in place of π_{j0} in the denominators, a factor $\xrightarrow{p} 1$; so $X^2 = \mathbf{e}_n^\top \mathbf{e}_n + o_p(1)$.

Step 6. $(I - H)\mathbf{z}$ has covariance $(I - H)(I - \mathbf{s}\mathbf{s}^\top)(I - H) = I - H - \mathbf{s}\mathbf{s}^\top$, as $H\mathbf{s} = \mathbf{0}$. H and $\mathbf{s}\mathbf{s}^\top$ project onto orthogonal subspaces of dimensions p and 1, so $I - H - \mathbf{s}\mathbf{s}^\top$ is idempotent of rank $c - p - 1$; Lemma 1.9(b) gives $X^2 \xrightarrow{d} \chi^2_{c-1-p}$.

Step 7. Step 4 gives $(n_j - \hat{\mu}_j)/\hat{\mu}_j = O_p(n^{-1/2})$ and $\sum_j \hat{\mu}_j = n$, so Theorem 1.17 gives $G^2 - X^2 \xrightarrow{p} 0$. ■

Example 1.19 (Pneumonia infections in calves)

156 calves were classified by primary infection and, if infected, secondary infection; (no primary, yes secondary) is a *structural zero*. Counts (fitted): (yes,yes) 30 (38.1), (yes,no) 63 (39.0), (no,no) 63 (78.9). If $P(\text{primary}) = \pi = P(\text{secondary} \mid \text{primary})$, the cell probabilities are $(\pi^2, \pi(1-\pi), 1-\pi)$, the kernel is $\pi^{2n_{11}+n_{12}}(1-\pi)^{n_{12}+n_{22}}$, and

$$\hat{\pi} = \frac{2n_{11} + n_{12}}{2n_{11} + 2n_{12} + n_{22}} = \frac{123}{249} = 0.494.$$

$X^2 = 19.7$, df $= 3 - 1 - 1 = 1$ ($P < 0.00001$): primary infection seems to have an immunizing effect.